

AI Assistance Note — Assignment 05

AI Platform Used

- Tool: Claude
- Date: August 2026

How AI Assisted My Work

Task	AI Assistance	Independent Work
Task 1 — Concept write-up	Helped structure sections and locate citation candidates	Selected the recommendation rationale, wrote the comparison table, reviewed all content for accuracy
Task 2 — anomaly_detector.py	Generated the CLI skeleton (train/predict/tune modes), the feature-alias mapping between UNSW-NB15 and my synthetic dataset's column names	Ran training end-to-end, verified feature selection and scaling were correct, checked model/scaler outputs
Task 3 — evaluation.py / evasion_test.py	Drafted the confusion-matrix plotting code and HTML report template	Ran the evaluation, interpreted the robustness results, wrote the ROBUSTNESS.md analysis and operational recommendations
Threshold tuning	Assisted with the precision/recall/F1 sweep loop	Reviewed the trade-offs and confirmed threshold=0.0 as the F1-optimal choice for this run

Prompts / Questions Asked

1. "Build a full Isolation Forest anomaly detection CLI with train/predict/tune modes following this assignment spec."
2. "How should the confusion matrix and evasion test be structured so recall/robustness numbers are computed correctly?"
3. "Help me diagnose why my per-attack-type breakdown numbers didn't add up to the overall recall."

Challenges Encountered

- Index misalignment bug: an early version of the train/test split reset the DataFrame index, which silently broke the per-attack-type breakdown (it was joining predictions to the wrong rows of the raw dataset). Caught this by noticing the per-attack rates didn't reconcile with the overall recall, then fixed the split to preserve original row indices.
- Feature scale mismatch: the assignment's reference features (dur, pkts, dpkts, sbytes, dbytes, rate) don't exist under those exact names in my Assignment 04 synthetic dataset, so I built an explicit alias map rather than renaming the dataset itself.

- Low robustness score: my first read of the evasion results assumed a bug, since I expected the model to be more robust; re-checking the math confirmed the low score is a genuine property of a 6-feature Isolation Forest at this threshold, not a computation error, and I documented it honestly in ROBUSTNESS.md instead of tuning the test to look better.

Lessons Learned

- Unsupervised models are still evaluated with supervised metrics here (precision/recall/F1) using held-out labels — the labels must never leak into training, only into evaluation.
- A model can have high precision and still be fragile to evasion; these are different properties and both need to be reported.
- Index handling in pandas is a common, quiet source of evaluation bugs — worth sanity-checking that per-subgroup metrics reconcile with the overall metric before trusting them.

Tasks Completed Independently

- Dataset feature mapping and selection
- Running the full train → tune → evaluate pipeline end-to-end
- Diagnosing and fixing the index-alignment bug in the evaluation pipeline
- Interpreting all metrics, robustness results, and writing the operational recommendations
- Final review of all submission files

This note fulfills the assignment requirement to disclose AI assistance transparently.